

Jiacheng Lai

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RESEARCH PROFILE

Undergraduate student in Mathematics and Applied Mathematics at Changsha University, with research experience in bioinformatics, computational biology, deep learning, and statistical modeling. My current work focuses on AI-driven drug-microbe association prediction and drug-target affinity prediction, with an emphasis on multimodal biological representation learning, graph neural networks, attention mechanisms, and interpretable models for drug repurposing. My undergraduate research has resulted in one first-author SCI-indexed journal article, one first-author journal manuscript under review, one co-first-author conference submission, and several open-source implementations of computational biology models.

RESEARCH INTERESTS

Bioinformatics; Computational Biology; Deep Learning; Artificial Intelligence; Statistical Modeling; Drug-Microbe Association Prediction; Drug-Target Affinity Prediction; AI for Drug Discovery

EDUCATION

Changsha University, Changsha, China

B.S. in Mathematics and Applied Mathematics, School of Mathematics

Sept. 2023 - June 2027 expected

GPA: 3.41/4.5; Ranking: Top 10%

Advisor: Prof. Lei Wang

Relevant Coursework: Mathematical Analysis, Advanced Algebra, Complex Analysis, Ordinary Differential Equations, Multivariate Statistics

PUBLICATIONS

Peer-Reviewed Journal Articles

Lai, J., Zhang, Z., Zeng, B., & Wang, L. (2025). GTFKAN: A novel microbe-drug association prediction model based on graph transformer and Fourier Kolmogorov-Arnold networks. *Journal of Molecular Biology*, 437(17), 169201.

<https://doi.org/10.1016/j.jmb.2025.169201>

Role: First author

Manuscripts Under Review

Lai, J., Zhang, Z., Yuan, H., & Wang, L. BMFN: Multimodal feature fusion network for drug-Mtb affinity prediction and anti-tuberculosis drug repurposing. Under review at *Bioinformatics*. Submitted May 25, 2026.

Role: First author

Code: <https://github.com/zjdssljc/BMFN>

Lai, J., Xu, X., Zhang, Z., Yuan, H., & Wang, L. MiDSFN: A novel signature fusion learning network for potential drug-microbe associations prediction. Submitted to *IEEE BIBM 2026*. Submitted June 4, 2026.

Role: Co-first author; listed first among equal contributors

Code: <https://github.com/MagicWings49/MiDSFN/>

RESEARCH EXPERIENCE

Research Assistant

Bioinformatics Complex Network Big Data Research Institute / Big Data Innovation and Entrepreneurship Education
Center of Hunan Province, Changsha University

Oct. 2023 - Present

- Conduct research on computational biology models for drug-microbe association prediction and drug-target affinity prediction.
- Develop deep learning models using Python and PyTorch, with applications involving heterogeneous biological networks, SMILES strings, 16S rDNA sequences, molecular fingerprints, and protein sequence data.
- Process public biological and chemical datasets including MDAD, aBiofilm, ChEMBL, Davis, and KIBA for model training, validation, and case studies.
- Use RDKit, PubChem API, Matplotlib, Scikit-learn, and PyMOL for molecular feature extraction, scientific visualization, model evaluation, and structural interpretation.

Project Leader, National College Student Innovation and Entrepreneurship Training Program

Project: A Novel Combined Neural Network Model for Predicting Microbe-Drug Associations Based on Bilinear Attention Networks

Project No.: S202411077005

May 2024 - Mar. 2026

- Led a national-level undergraduate innovation project on deep learning-based microbe-drug association prediction.
- Responsible for overall project design, model development, experimental analysis, manuscript writing, and team coordination.
- Developed a series of computational biology models, including approaches based on SVD, Node2vec, graph neural networks, graph transformers, Fourier Kolmogorov-Arnold networks, and bilinear attention networks.
- Advanced the project from topology-dependent prediction toward intrinsic biological feature-driven modeling using drug SMILES strings and microbial 16S rDNA sequences.
- Contributed to research outputs including a first-author article in **Journal of Molecular Biology** and a co-first-author submission to IEEE BIBM 2026.

Core Developer, Microbe-Drug Association Prediction Under Complex Networks

Changsha University

Oct. 2023 - Present

- Designed and implemented GTFKAN, a model combining graph transformer networks and Fourier Kolmogorov-Arnold networks for potential microbe-drug association prediction.
- Constructed heterogeneous biological networks and extracted structural, positional, and higher-order latent features for microbe and drug representations.
- Performed ablation studies, comparative experiments, and case studies to evaluate model performance against state-of-the-art baselines.
- Released open-source implementation: <https://github.com/zjdssljc/GTFKAN>

Project Leader, Multimodal Feature Fusion for Anti-Tuberculosis Drug-Target Affinity Prediction

Changsha University

May 2025 - Present

- Designed BMFN, a multimodal feature fusion network for drug-Mtb target affinity prediction and anti-tuberculosis drug repurposing.
- Integrated molecular graphs, protein sequences, molecular fingerprints, channel-spatial attention, graph learning, and sequence encoding for pathogen-specific DTA prediction.
- Processed Mtb-specific datasets from ChEMBL and related sources, involving over 17,000 compounds, and evaluated model performance on both Mtb-specific and general DTA benchmarks.
- Conducted ablation experiments, model interpretation, and molecular docking-based validation for drug repurposing analysis.
- Released open-source implementation: <https://github.com/zjdssljc/BMFN>

Research Contributor, NSFC Proposal Preparation

Project: Intelligent Anti-Tuberculosis Drug Screening Model and Algorithm Based on Semi-Evolutionary Attention Mechanism

Changsha University

Sept. 2025 - Feb. 2026

- Contributed to the preparation of a National Natural Science Foundation of China proposal on AI-driven anti-tuberculosis drug screening.
- Assisted in writing the literature review, research background, scientific rationale, and domestic and international research progress sections.
- Supported the design of technical routes involving semi-evolutionary attention mechanisms, multimodal biological feature fusion, and interpretable deep learning models for Mtb drug screening.

SELECTED PRESENTATIONS

Oral Presentation, The 16th "Challenge Cup" Hunan Provincial College Student Extracurricular Academic Science and Technology Works Competition

Title: Zhilian Weiyao: Efficient Deep Learning Architecture Design and Application for Microbe-Drug Association Prediction

June 2025

Project Proposal Defense, National College Student Innovation and Entrepreneurship Training Program

Title: A Novel Combined Neural Network Model for Predicting Microbe-Drug Associations Based on Bilinear Attention Networks

May 2024

Oral Defense, Zhengda Cup National College Student Market Survey and Analysis Competition

Title: Preserving the Millennium Legacy of Miao Costumes: A Survey on Changsha College Students' Awareness of Miao Ethnic Clothing and Its Influencing Factors

Apr. 2025

MENTORING AND ACADEMIC SERVICE

Undergraduate Research Mentor

Bioinformatics Complex Network Big Data Research Institute, Changsha University

2024 - Present

- Mentor junior undergraduate students in literature review, research project design, computational biology workflows, and project implementation.
- Lead internal code training sessions on Python, PyTorch, model evaluation, and reproducible research practices.
- Organize and participate in group discussions on recent papers in bioinformatics, graph neural networks, multimodal learning, and AI for Science.

HONORS AND AWARDS

Third Prize, The 16th "Challenge Cup" Hunan Provincial College Student Extracurricular Academic Science and Technology Works Competition, June 2025

Second Prize, Zhengda Cup National College Student Market Survey and Analysis Competition, Apr. 2025

Honorable Mention, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM), COMAP, Feb. 2025

Project Leader, National College Student Innovation and Entrepreneurship Training Program, Nov. 2024

Second Prize, National College Student Mathematics Competition, Category B, Dec. 2024

Second Prize, Hunan Provincial College Student Mathematics Competition, Category B, Dec. 2024

Third Prize, Hunan Provincial College Student Market Survey and Analysis Competition, Aug. 2025

Third Prize, Changsha College Student Data Element-Empowered Artificial Intelligence Innovation and Entrepreneurship Competition, Dec. 2025

TECHNICAL SKILLS

Programming and Frameworks: Python, PyTorch, Scikit-learn, object-oriented programming, reproducible research workflows

Bioinformatics and Computational Biology: RDKit, PubChem API, SMILES, 16S rDNA sequences, molecular fingerprints, protein sequences, heterogeneous biological networks

Deep Learning and Machine Learning: Graph neural networks, graph transformers, bilinear attention networks, self-attention mechanisms, residual neural networks, multimodal feature fusion, random forest, XGBoost

Data Analysis and Visualization: Matplotlib, scientific plotting, model evaluation, ablation studies, case studies

Molecular Visualization and Interpretation: PyMOL, protein structure visualization, molecular docking interpretation

Mathematical Background: Mathematical analysis, linear algebra, complex analysis, ordinary differential equations, multivariate statistics

SELECTED RESEARCH CODE

BMFN: <https://github.com/zjdssljc/BMFN>

GTFKAN: <https://github.com/zjdssljc/GTFKAN>

MiDSFN: <https://github.com/MagicWings49/MiDSFN/>